

Gennoor Tech

TRAIN. INNOVATE. BUILD.

GENNOOR ACADEMY · SELF-PACED ·
70 MIN

AI in Manufacturing

A 70-minute playbook for plant managers and VPs of manufacturing — six plant-floor plays that ship, three pitches that disappoint, the OT/IT bridge filter, and a 12-month plant roadmap with OEE and quality KPIs.

8

CHAPTERS

70 min

DURATION

Intermediate

LEVEL

FREE

CERTIFICATE

Course Overview

No vendor-pitch hype. Six manufacturing AI plays that consistently survive the shop floor — predictive maintenance, quality vision, process optimisation, plant-floor supply chain, safety AI, and pragmatic digital twin — and three pitches that consistently disappoint when deployed in real plants, named explicitly. The OT/IT bridge filter that sequences your specific plant.

FORMAT: Self-paced · 8 chapters · audio-narrated slides · interactive end-of-course builder · free certificate on completion

Who Is This For?

1

Plant managers · VPs
Manufacturing

2

Operations leadership

3

Process + quality
engineers

4

IT/OT leaders

Learning Outcomes

- 1 6 plant-floor plays that ship — predictive maintenance, vision quality, energy, scheduling, safety, throughput
- 2 3 pitches that disappoint — and the OT/IT bridge filter that separates them
- 3 How to design the OT/IT bridge that makes AI on the shop floor possible
- 4 The digital-twin reality applied honestly — what works, what's marketing, what breaks at production scale
- 5 The ISA-95 stack as the deployment frame — where AI plugs in safely vs where it threatens uptime
- 6 A 12-month plant roadmap with OEE and quality KPIs as the success measure
- 7 The trust trip-wires — worker safety, equipment integrity, regulatory disclosure

8-Chapter Curriculum

Self-paced, audio-narrated slide format. Each chapter ends with a "Monday move" the learner can apply that week.

01

The manufacturing-AI fit map

CHAPTER 1 · 9 MIN · NARRATED + INTERACTIVE

- Six plant-floor plays that ship
- three pitches that disappoint on the shop floor
- the OT/IT bridge filter (data accessible
- data quality usable
- culture supports AI).

02

Predictive maintenance for plants

CHAPTER 2 · 10 MIN · NARRATED + INTERACTIVE

- Three sensor coverage decisions
- the narrative layer the engineer acts on
- three project killers most plants hit during deployment.

03

Quality vision AI

CHAPTER 3 · 10 MIN · NARRATED + INTERACTIVE

- Three messy-real-world challenges (lighting
- drift
- novel defects)
- three-stage operator labelling loop
- the false-confidence year-2 failure mode + three disciplines.

04

Process optimisation with AI

CHAPTER 4 · 9 MIN · NARRATED + INTERACTIVE

- Three signals AI surfaces (multi-parameter interactions
- drift
- best-shift patterns)
- operator-led experiment design
- three guards against the overfitting trap.

05

Plant-floor supply chain integration

CHAPTER 5 · 9 MIN · NARRATED + INTERACTIVE

- Three signal flows into scheduling (demand
- material/supplier
- equipment availability)
- three re-planning cadences
- three failure modes when integration breaks.

06

Safety AI — augmenting human attention

CHAPTER 6 · 9 MIN · NARRATED + INTERACTIVE

- Three safety use cases (PPE
- near-miss
- ergonomic)
- three failure modes specific to safety
- the one line nobody crosses + the safety AI charter.

07

Digital twin — pragmatic, not megaproject

CHAPTER 7 · 9 MIN · NARRATED + INTERACTIVE

- Three targeted twins that ship in 4–6 months
- three megaproject traps to refuse
- three discipline rules (6-month time-to-value
- one bounded use case
- real plant engineers in design).

08

Capstone — Your 12-month plant AI roadmap

CHAPTER 8 · 8 MIN · NARRATED + INTERACTIVE

- Two plays Q1 (PdM + quality vision), ranging Q2–Q4
- OEE and quality KPIs not AI metrics
- four conversations Monday.